Md. Tarek Hasan

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Education

January '24 – August '25 : **Master of Science in Computer Science and Engineering**, *Department of Computer Science and Engineering, United International University, United City, Madani Ave, Dhaka 1212.*
CGPA: 4.00 out of 4.00, **Gold Medalist** of the 8th Convocation of United International University

February '18 – February '22 : **Bachelor of Science in Computer Science and Engineering**, *Department of Computer Science and Engineering, United International University, United City, Madani Ave, Dhaka 1212.*
CGPA: 4.00 out of 4.00, **Valedictorian & Gold Medalist** of the 7th Convocation of United International University

Research Experience

June 2024 – September 2024 **Summer Research Intern**, School of Computing, National University of Singapore.
Advisor: Assoc. Prof. Terence Sim; mentors: Dr. Sanjay Saha and Dr. Sanka Rasnayaka.

- **One-shot biometric authentication:** Proposed *GenAI Augmented Enrollment (GAE)*, which uses a Stable Diffusion–based face animation model (AniFaceDiff, with ReferenceNet identity conditioning) to synthesize pose- and expression-varied enrollment samples from a single face image, then averages their embeddings into a robust biometric signature.
- Benchmarked GAE against single-sample enrollment across six face recognition models (ArcFace, FaceNet, FaceNet512, GhostFaceNet, SFace, VGG-Face) on CASIA-WebFace, LFW, and VoxCeleb, reducing Equal Error Rate (EER) by **25.98%** on average, with the largest gains for non-frontal enrollment.
- **Deepfake synthesis vs. detection:** Generated talking-head deepfake videos with GAN-, diffusion-, and NeRF-based methods (e.g., TPSMM, HyperReenact, FADM, AniFaceDiff, SyncTalk) and benchmarked state-of-the-art detectors (Xception, CORE, RECCE, UCF, SRM, Capsule, and others) using AUC and AP, exposing poor generalization to unseen, diffusion-based generators.
- Designed and ran a human perception study on the same video stimuli, with audio removed and a six-point confidence scale converted to probabilities, enabling a direct human-vs.-detector comparison; this work became the arXiv preprint *Deepfake Synthesis vs. Detection: An Uneven Contest* (2026).

Selected Publications

In Conference Proceedings

2026 Many dialects, many languages, one cultural lens: Evaluating multilingual VLMs for Bengali culture understanding across historically linked languages and regional dialects. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, Budapest, Hungary, 2026. Association for Computational Linguistics.

2022 **Md. Tarek Hasan**, Arifa Akter, Mohammad Nazmush Shamael, Md Al Emran Hossain, HM Mutasim Billah, Sumayra Islam, and Swakkhar Shatabda. Adaptive tabu dropout for regularization of deep neural networks. In *International Conference on Neural Information Processing*, pages 355–366. Springer, 2022, doi:10.1007/978-3-031-30105-6$_3$0.

Journal Articles

2026 **Md. Tarek Hasan**, Faiyaz Rayhan, Myisha Tun Samiha, Minhajul Bashir, Md Sakhawat Hossain, Suman Ahmmed, and Ohidujjaman. Architecture-oriented benchmarking and weighted ensemble learning for multi-class plant disease classification. *Array*, page 101348. Elsevier, 2026, doi:10.1016/j.array.2026.101348.

2026 Debopom Sutradhar, Ripon Debnath, Subhey Rahman, **Md. Tarek Hasan**, and Sami Azam. Dpmt-net: a dual-prompt attention multi-task 3d network for joint nodule segmentation and malignancy prediction in lung ct scans. *Complex & Intelligent Systems*. Springer, 2026, doi:10.1007/s40747-026-02337-w.

2026 Ken Chen, Sachith Seneviratne, Wei Wang, Dongting Hu, Sanjay Saha, **Md. Tarek Hasan**, Sanka Rasnayaka, Tamasha Malepathirana, Mingming Gong, and Saman Halgamuge. Anifacediff: Animating stylized avatars via parametric conditioned diffusion models. *Pattern Recognition*, volume 170, page 112017, 2026, doi:10.1016/j.patcog.2025.112017.

2023 **Md. Tarek Hasan**, Md. Al Emran Hossain, Md. Saddam Hossain Mukta, Arifa Akter, Mohiuddin Ahmed, and Salekul Islam. A Review on Deep-Learning-Based Cyberbullying Detection. *Future Internet*, volume 15, 2023, doi:10.3390/fi15050179.

Professional Experience

February 2026 – Present **Assistant Professor and Undergraduate Program Coordinator**, Department of Computer Science and Engineering, United International University.
Deliver core computing courses and mentor students while overseeing undergraduate curriculum planning, course coordination, and academic operations. Ensure compliance with institutional policies and accreditation requirements, provide academic advising, and contribute to admissions, quality assurance, and program development.

June 2022 – February 2026 **Lecturer**, Department of Computer Science and Engineering, United International University.
Taught courses including Artificial Intelligence, Structured Programming, Object-Oriented Programming, and Data Structures and Algorithms. Served as **Moderator of the UIU Computer Club**, supervised student theses, facilitated competitive programming, and contributed to administrative and accreditation activities.

February 2022 – June 2022 **Software Engineer**, Samsung R&D Institute Bangladesh, Samsung Research.
Developed and optimized features for the SmartThings iOS platform, focusing on app functionality, IoT integration, and user experience. Applied Swift programming and Scrum methodology in a collaborative development environment.

Competitive Programming

Platforms Codeforces Specialist (max rating 1414; best rank 814th globally); LeetCode contest rating 1818 (674 problems solved; best rank 254); Google Code Jam 2021 Round 1 qualifier.

Contests Problem Setter, Judge, and Boot Camp Instructor, UIU Coders Combat 3.0; Ranked 4th, UIU Coders Combat 2.0.

Honors & Awards

Academic Valedictorian and Chancellor's Gold Medalist (B.Sc., 7th Convocation); Gold Medalist (M.Sc., 8th Convocation); both degrees Summa Cum Laude, with a 100% merit scholarship in every trimester at UIU.

Research Dr. Fatema Rashid Best Paper Award (2nd Place), ECCE 2023.

Competitions Theme Category Winner, Blockchain Olympiad Bangladesh 2021, and national representative at the International Blockchain Olympiad 2021 (e-Health); Champion, Software Engineering Project, UIU CSE Project Show (Fall 2021).